TSAR Project Part 2 – Data Wrangling with dplyr and tidyr

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1 Task 1

1.1 Data Loaded

```
myLCdata <- fread("myLCdata.csv")
```

1.2 Data dirtyfied

```
# Dirtyfy data and store it so we work with the same dirtyfy data
# dirtyfy <- readRDS("dirtyfy.rds")
# myLCdata_dirty <- dirtyfy(myLCdata, 3)
# fwrite(myLCdata_dirty, file = "myLCdata_dirty.csv")
myLCdata_dirty <- fread("myLCdata_dirty.csv")
```

1.3 Automatic EDA report

The following command was used to generate an automatic EDA report for `myLCdata_dirty`. As required, it was executed in the R console and is shown here only for documentation purposes.

```
create_report(myLCdata_dirty, output_file = "EDA_report_P2_03.html")
```

1.4 Missing Data Profile

Figure 1 shows the Missing Data Profile generated from the dirty private data set.

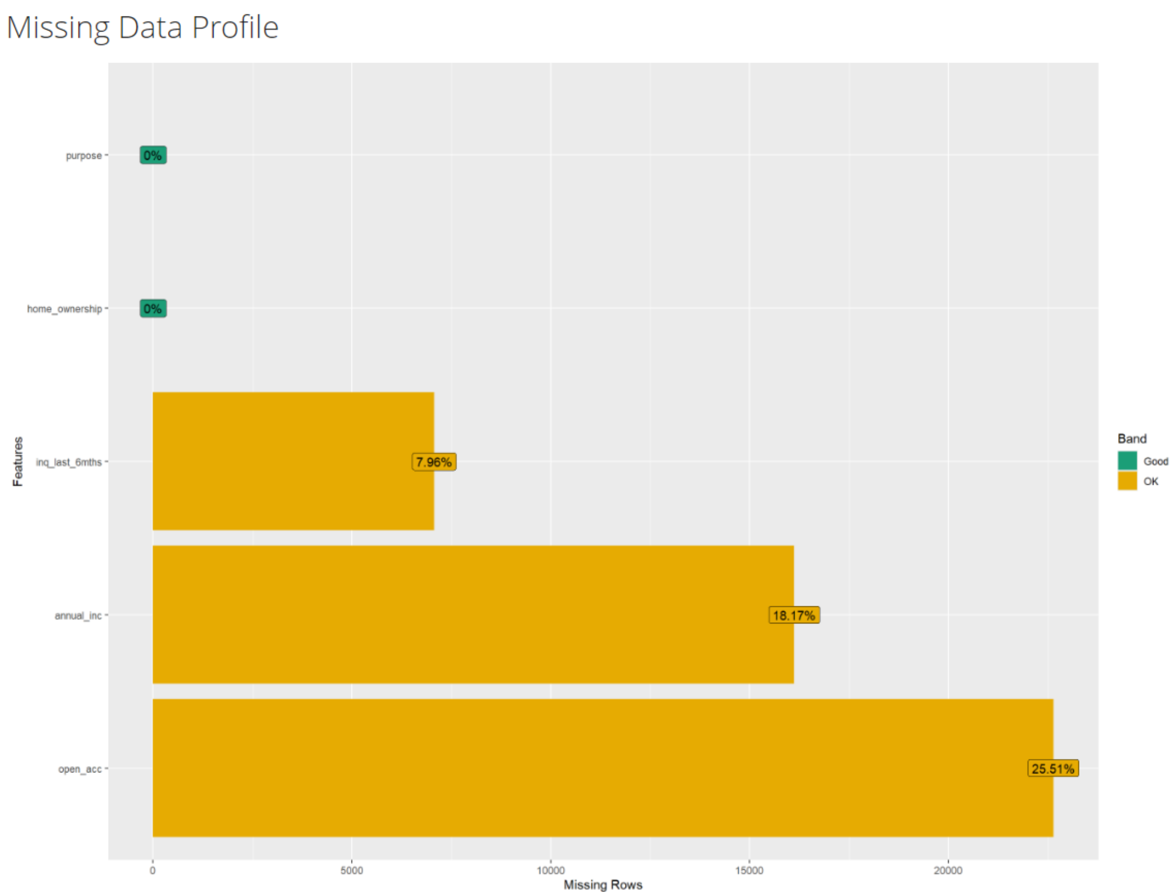


Figure 1: Missing Data Profile of the dirty private data set.

2 Task 2

2.1 Special values in character attributes

```
summary(myLCdata_dirty)
```

```
 home_ownership        annual_inc         purpose           inq_last_6mths
 Length:88738       Min.   :   3000    Length:88738       Min.   : 0.000
```

```
 Class :character   1st Qu.:  45000   Class :character   1st Qu.: 0.000  
 Mode  :character   Median :  64500   Mode  :character   Median : 0.000  
                    Mean   :  75147                      Mean   : 0.693  
                    3rd Qu.:  90000                      3rd Qu.: 1.000  
                    Max.   :6000000                      Max.   :31.000  
                    NA's   :16120                        NA's   :7067    
    open_acc    
 Min.   : 0.00  
 1st Qu.: 8.00  
 Median :11.00  
 Mean   :11.52  
 3rd Qu.:14.00  
 Max.   :76.00  
 NA's   :22640  
```

First We count special values for character features ### All columns #### Number of NA

```
num_na <- myLCdata_dirty %>% summarize(across(everything(), ~sum(is.na(.))))
num_na
```

```
  home_ownership annual_inc purpose inq_last_6mths open_acc
1              0      16120       0           7067    22640
```

2.1.1 Numeric Columns

2.1.1.1 Number of NaN

```
num_nan <- myLCdata_dirty %>% summarize(across(where(is.numeric), ~sum(is.nan(.))))
num_nan
```

```
  annual_inc inq_last_6mths open_acc
1          0              0        0
```

2.1.1.2 Number of Outliers

```
# Normal way
num_outliers <- myLCdata_dirty %>% summarize(across(where(is.numeric), ~length(boxplot.stats
num_outliers
```

```
  annual_inc inq_last_6mths open_acc
1       3315           4948     2018
```

```
# Fancy way; Same output
# myLCdata_dirty %>% summarize(across(where(is.numeric), ~sum(
#     boxplot.stats(.)$stats[1] > . | . > boxplot.stats(.)$stats[5],
#     na.rm = TRUE
#   )))
```

2.1.2 Character Columns

2.1.2.1 Special Strings "n/a"

```
num_na_string <- myLCdata_dirty %>% summarize(across(where(is.character), ~sum(
    . == "n/a"
  )))
num_na_string
```

```
  home_ownership purpose
1           6301    7941
```

2.1.2.2 Special Strings Space " "

```
num_white_space <- myLCdata_dirty %>% summarize(across(where(is.character), ~sum(
    . == " "
  )))
num_white_space
```

```
  home_ownership purpose
1              0       0
```

2.1.2.3 Special Strings Empty String ""

```
num_empty_string <- myLCdata_dirty %>% summarize(across(where(is.character), ~sum(
    "" == .
  )))
num_empty_string
```

```
  home_ownership purpose
1          15327   18339
```

##Task2.2

```r
#total number of rows
total_rows = sapply(myLCdata_dirty, length)

#summing up the special values

total_special = bind_rows(num_na, num_nan, num_outliers, num_na_string, num_white_space, num_
  summarise(across(everything(), ~sum(., na.rm = TRUE)))

relative_counts = total_special / total_rows

relative_counts
```

```
  home_ownership annual_inc   purpose inq_last_6mths  open_acc
1      0.2437287  0.2190155 0.2961527      0.1353986 0.2778742
```

##Task 2.3

###numeric attributes

```r
# Total rows
num_total = nrow(myLCdata_dirty)

# numeric_na metric
numeric_na = myLCdata_dirty %>% summarize(across(where(is.numeric), ~sum(is.na(.))))

# table
numeric_result_table = data.frame(
  attribute     = names(numeric_na),
  perc_NA       = as.numeric(numeric_na / num_total),
  perc_NaN      = as.numeric(num_nan / num_total),
  perc_outliers = as.numeric(num_outliers / num_total)
)

numeric_result_table
```

```
       attribute    perc_NA perc_NaN perc_outliers
```

```
1     annual_inc 0.18165837        0    0.03735716
2 inq_last_6mths 0.07963894        0    0.05575965
3       open_acc 0.25513309        0    0.02274110
```

###character attribute

```
# character_na metric
character_na = myLCdata_dirty %>% summarize(across(where(is.character), ~sum(is.na(.))))

#character table
character_result_table = data.frame(
  attribute = names(character_na),
  perc_NA = as.numeric(character_na / num_total),
  perc_empty_string = as.numeric(num_empty_string / num_total),
  perc_space = as.numeric(num_white_space / num_total),
  perc_na_string = as.numeric(num_na_string / num_total)
)

character_result_table
```

```
       attribute perc_NA perc_empty_string perc_space perc_na_string
1 home_ownership       0         0.1727219          0     0.07100678
2        purpose       0         0.2066646          0     0.08948816
```

##Task 2.4

```
#numeric tidy table

tidy_numeric_result_table = numeric_result_table %>%
  pivot_longer(
    cols = starts_with("perc_"),
    names_to = "metric",
    values_to = "value"
  )
tidy_numeric_result_table
```

```
# A tibble: 9 x 3
  attribute      metric          value
  <chr>          <chr>           <dbl>
1 annual_inc     perc_NA       0.182
2 annual_inc     perc_NaN      0
```

```
3 annual_inc     perc_outliers 0.0374
4 inq_last_6mths perc_NA       0.0796
5 inq_last_6mths perc_NaN      0
6 inq_last_6mths perc_outliers 0.0558
7 open_acc       perc_NA       0.255
8 open_acc       perc_NaN      0
9 open_acc       perc_outliers 0.0227
```

```r
#character tidy table

tidy_character_result_table = character_result_table %>%
  pivot_longer(
    cols = starts_with("perc_"),
    names_to = "metric",
    values_to = "value"
  )

tidy_character_result_table
```

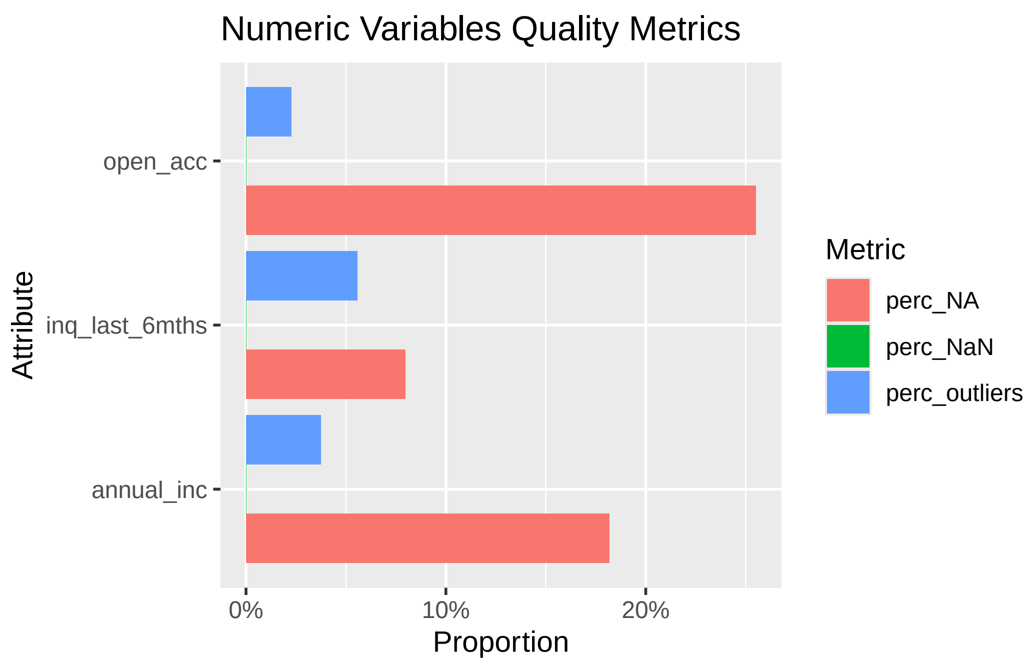
```
# A tibble: 8 x 3
  attribute      metric             value
  <chr>          <chr>              <dbl>
1 home_ownership perc_NA           0
2 home_ownership perc_empty_string 0.173
3 home_ownership perc_space        0
4 home_ownership perc_na_string    0.0710
5 purpose        perc_NA           0
6 purpose        perc_empty_string 0.207
7 purpose        perc_space        0
8 purpose        perc_na_string    0.0895
```

##Task 2.5

###numeric table

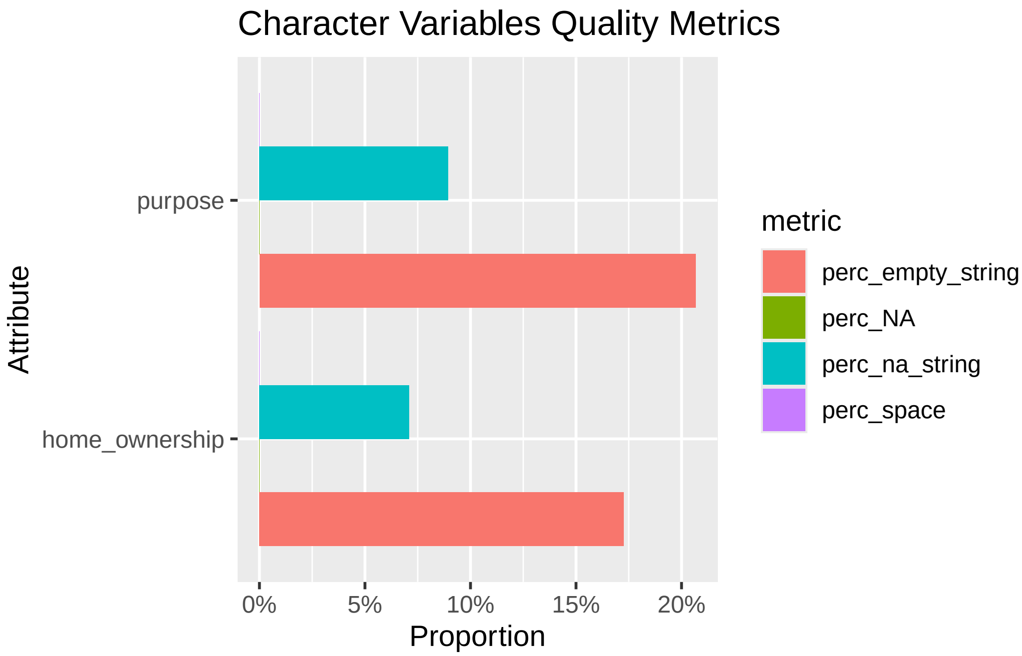
```r
ggplot(tidy_numeric_result_table, aes(x = attribute, y = value, fill = metric)) +
  geom_col(position = "dodge") +
  labs(
    title = "Numeric Variables Quality Metrics",
    x = "Attribute",
    y = "Proportion",
    fill = "Metric"
```

```
  ) +
  scale_y_continuous(labels = scales::percent)+
  coord_flip()
```



###character table

```
ggplot(tidy_character_result_table, aes(x= attribute, y = value, fill = metric))+
  geom_col(position = "dodge")+
  labs(
    title = "Character Variables Quality Metrics",
    x = "Attribute",
    y = "Proportion"
    )+
  coord_flip()+
  scale_y_continuous(labels = scales::percent)
```


Character Variables Quality Metrics
purpose
home_ownership
Attribute
0%
5%
10%
15%
20%
Proportion
metric
perc_empty_string
perc_NA
perc_na_string
perc_space